

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=pd.read_csv("customer_shopping_behavior.csv")
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)
0	1	55	Male	Blouse	Clothing	53
1	2	19	Male	Sweater	Clothing	64
2	3	50	Male	Jeans	Clothing	73
3	4	21	Male	Sandals	Footwear	90
4	5	45	Male	Blouse	Clothing	49

	Location	Size	Color	Season	Review Rating	Subscription Status
0	Kentucky	L	Gray	Winter	3.1	Yes
1	Maine	L	Maroon	Winter	3.1	Yes
2	Massachusetts	S	Maroon	Spring	3.1	Yes
3	Rhode Island	M	Maroon	Spring	3.5	Yes
4	Oregon	M	Turquoise	Spring	2.7	Yes

	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	Express	Yes	Yes	14
1	Express	Yes	Yes	2
2	Free Shipping	Yes	Yes	23
3	Next Day Air	Yes	Yes	49
4	Free Shipping	Yes	Yes	31

	Payment Method	Frequency of Purchases
0	Venmo	Fortnightly
1	Cash	Fortnightly

2	Credit Card	Weekly
3	PayPal	Weekly
4	PayPal	Annually

```
df.isnull().sum()
```

Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	37
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Payment Method	0
Frequency of Purchases	0
dtype: int64	

```
df["Review Rating"]=df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
df.isnull().sum()
```

Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	0
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Payment Method	0
Frequency of Purchases	0
dtype: int64	

```
df.columns=df.columns.str.lower()
df.columns=df.columns.str.replace(' ','_')
df.columns
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount_(usd)', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

Create New age column

```
labels=['Young Adult','Adult','Middle-aged','Senior']
df['age_group']=pd.qcut(df['age'],q=4,labels=labels)
df[['age','age_group']].head(10)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

#create column purchase_frequency_days

```
df.head()
```

	customer_id	age	gender	item_purchased	category
0					
53					
1					
64					
2					
73					
3					
90					
4					
49					
	location	size	color	season	review_rating

subscription_status	\				
0	Kentucky	L	Gray	Winter	3.1
Yes					
1	Maine	L	Maroon	Winter	3.1
Yes					
2	Massachusetts	S	Maroon	Spring	3.1
Yes					
3	Rhode Island	M	Maroon	Spring	3.5
Yes					
4	Oregon	M	Turquoise	Spring	2.7
Yes					

	shipping_type	discount_applied	promo_code_used	previous_purchases
0	Express	Yes	Yes	14
1	Express	Yes	Yes	2
2	Free Shipping	Yes	Yes	23
3	Next Day Air	Yes	Yes	49
4	Free Shipping	Yes	Yes	31

	payment_method	frequency_of_purchases	age_group
0	Venmo	Fortnightly	Middle-aged
1	Cash	Fortnightly	Young Adult
2	Credit Card	Weekly	Middle-aged
3	PayPal	Weekly	Young Adult
4	PayPal	Annually	Middle-aged

```
frequency_mapping={
    'Fortnightly':14,
    'Weekly':7,
    'Monthly':30,
    'Quarterly':90,
    'Bi-weekly':14,
    'Annually':365,
    'Every 3 Month':90,
}
df['purchase_frequency_days']=df['frequency_of_purchases'].map(frequency_mapping)
```

```
df[['frequency_of_purchases','purchase_frequency_days']].head()
```

	frequency_of_purchases	purchase_frequency_days
0	Fortnightly	14.0
1	Fortnightly	14.0
2	Weekly	7.0

3	Weekly	7.0
4	Annually	365.0

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 3900 entries, 0 to 3899
```

```
Data columns (total 20 columns):
```

#	Column	Non-Null Count	Dtype
0	customer_id	3900 non-null	int64
1	age	3900 non-null	int64
2	gender	3900 non-null	object
3	item_purchased	3900 non-null	object
4	category	3900 non-null	object
5	purchase_amount_(usd)	3900 non-null	int64
6	location	3900 non-null	object
7	size	3900 non-null	object
8	color	3900 non-null	object
9	season	3900 non-null	object
10	review_rating	3900 non-null	float64
11	subscription_status	3900 non-null	object
12	shipping_type	3900 non-null	object
13	discount_applied	3900 non-null	object
14	promo_code_used	3900 non-null	object
15	previous_purchases	3900 non-null	int64
16	payment_method	3900 non-null	object
17	frequency_of_purchases	3900 non-null	object
18	age_group	3900 non-null	category
19	purchase_frequency_days	2769 non-null	float64

```
dtypes: category(1), float64(2), int64(4), object(13)
```

```
memory usage: 583.0+ KB
```

```
(df[['discount_applied', 'promo_code_used']]).all()
```

```
discount_applied    True
```

```
promo_code_used      True
```

```
dtype: bool
```

```
df=df.drop('promo_code_used',axis=1)
```

```
pip install mysql-connector-python
```

```
pip install sqlalchemy
```

```
from sqlalchemy import create_engine
```

```
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```

```
import urllib.parse
```

```

# 사용자 정보 설정
user = "root"
password = "Ashirbad@6372"

# '@' 문자를 URL 인코딩하여 비밀번호를 안전하게 인코딩
safe_password = urllib.parse.quote_plus(password)

# Engine 생성
engine = create_engine(f"mysql+mysqlconnector://{user}:{safe_password}@localhost/customer_behavior")

# 데이터프레임 데이터를 데이터베이스에 저장
df.to_sql(
    name='customer_data',
    con=engine,
    if_exists='replace',
    index=False
)

df

from sqlalchemy import create_engine
import urllib.parse

user = "root"
password = "Ashirbad@6372"
safe_password = urllib.parse.quote_plus(password)

# Engine 생성
engine = create_engine(f"mysql+mysqlconnector://{user}:{safe_password}@localhost/customer_behavior")

# 데이터베이스에 연결 시도
try:
    # 데이터베이스에 연결 시도
    with engine.connect() as connection:
        print("Success: Python successfully connected to MySQL!")
except Exception as e:
    print("Connection Failed!")
    print("Error Details:", e)

```